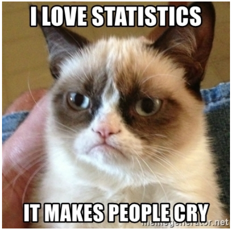
Cox Models for Political Science:

An Introduction to Duration (Survival) Analysis



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Outline

1. Introduction to Duration Analysis
2. Cox Proportional Hazard Model
3. Example 1: Duration in Authoritarian Regime
4. Example 2: Veterans' Administration Lung Cancer Dataset

Introduction to Duration Analysis

What is survival in Political Science?

- We want to study duration of political events or institutions.
- For example:
  - How long a regime survives before collapsing?
  - How long a cabinet remains in office?
  - How long a particular leader stays in power?

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	cowcode	year	gwf_country	gwf_casename	gwf_startdate	gwf_enddate	gwf_spell	gwf_duration	gwf_fail
	<int>	<int>	<chr>	<chr>	<chr>	<chr>	<int>	<int>	<int>
1	40	1953	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	1	0
2	40	1954	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	2	0
3	40	1955	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	3	0
4	40	1956	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	4	0
5	40	1957	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	5	0
6	40	1958	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	6	0
7	40	1959	Cuba	Cuba 52-59	1952-03-10	1959-01-01	7	7	1

Cox Proportional Hazard Model

Key Features of the Cox Model (1/2)

- The Cox model answers the question:
  - **Which variables (such as education, age, gender, etc.) affect the hazard rate?**

In other words…

The Cox model estimates how certain characteristics accelerate or decelerate the timing of an event's occurrence.

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- The hazard rate for the i-th individual is defined as

$$\begin{aligned} h_i(t) &= h_0(t)\exp(\beta' X_i) && (1) \\ &= h_0(t)\ \exp(\beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}) && (2) \end{aligned}$$

where $h_0(t)$ is the **baseline hazard**,
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> **! Important**
>
> In Cox model, the main *Quantity of Interest* we care about is *the hazard rate ($h_i(t)$).*

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- In *parametric* models, we apply **Maximum Likelihood Estimation (MLE)** to maximize the likelihood function and obtain the parameter estimates that best fit the data.
- In *semi-parametric* models, such as Cox model, we rely on *Partial* Likelihood Estimation, as we can only maximize part of the parameters, while leaving $h_0(t)$ is unspecified.

Example 1: Duration in Authoritarian Regime

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- **Treatment**:
  - `gwf_military`, coded as 1 if a country is classified as a military regime, otherwise 0.

```r
# library {survival}
library(survival)

# Fit cox model
model <- coxph(
  Surv(gwf_duration, gwf_fail) ~ gwf_military + wdi_gdpcapcur + wdi_pop,
  data = new_df
)
```

How to Read the Cox Model Output

```r
summary(model)
```

```
Call:
coxph(formula = Surv(gwf_duration, gwf_fail) ~ gwf_military +
    wdi_gdpcapcur + wdi_pop, data = new_df)

  n= 3141, number of events= 157
   (1450 observations deleted due to missingness)

                    coef  exp(coef)   se(coef)      z Pr(>|z|)
gwf_military   2.231e+00  9.307e+00  1.916e-01 11.642  < 2e-16 ***
wdi_gdpcapcur -5.076e-04  9.995e-01  9.224e-05 -5.503 3.73e-08 ***
wdi_pop       -7.013e-09  1.000e+00  3.263e-09 -2.149   0.0316 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

              exp(coef) exp(-coef) lower .95 upper .95
gwf_military     9.3072     0.1074    6.3932   13.5494
wdi_gdpcapcur    0.9995     1.0005    0.9993    0.9997
wdi_pop          1.0000     1.0000    1.0000    1.0000
```

Counterfactual Prediction (1/6)

```r
library(marginaleffects)

min_time <- min(new_df$gwf_duration[new_df$gwf_fail == 1], na.rm = TRUE)
max_time <- max(new_df$gwf_duration[new_df$gwf_fail == 1], na.rm = TRUE)

nd <- datagrid(
  gwf_military = 0:1,                                                        # 1
  gwf_duration = round(seq(min_time, max_time, length.out = 25)),           # 2
  grid_type = "counterfactual",                                              # 3
  model = model                                                              # 4
)
```

1. Specify groups to compare (military vs. non-military regimes)
2. 25 equally spaced points in time between the first (`min_time`) and the last (`max_time`) occurring event time
3. Tells the function to hypothesize specific values of predictors (`gwf_military` == 1 or 0), regardless of the observed values in the original dataset.
4. Specify the model used to generate predictions.

Counterfactual Prediction (2/6)

```r
p <- avg_predictions(model,
                     type = "survival",                          # 1
                     by = c("gwf_duration", "gwf_military"),     # 2
                     vcov = "rsample",                           # 3
                     newdata = nd)                               # 4
```

1. Choose the survival probability scale.
2. Calculate probability for each combination of `gwf_military` and `gwf_duration`. Every probability represents a point (survival probability).
3. Non-parametric bootstrapping.
4. Feed `avg_prediction()` with the new data that we just simulated.

Counterfactual Prediction (3/6)

```r
tail(p)
```

```
   gwf_duration gwf_military  estimate  conf.low conf.high
45           96            0 0.7365691 0.5827513 0.8514241
46           96            1 0.1708886 0.1009036 0.3331206
47          101            0 0.7365691 0.5827513 0.8514241
48          101            1 0.1708886 0.1009036 0.3331206
49          105            0 0.7365691 0.5827513 0.8514241
50          105            1 0.1708886 0.1009036 0.3331206
```

Counterfactual Prediction (4/6)

- It is useful to factor variables before plotting

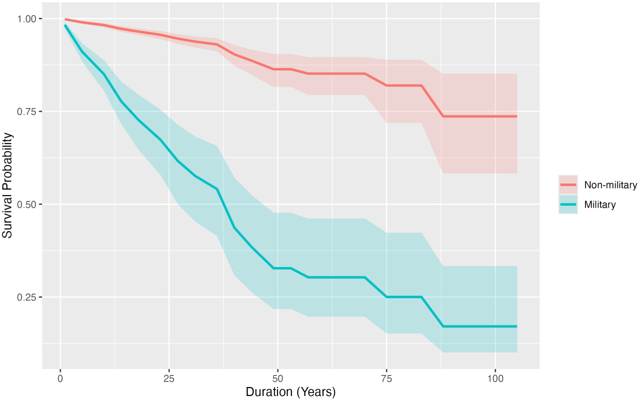
```
# Factoring treatment variable
# library(dplyr)
p <- p |>
  mutate(gwf_military = factor(gwf_military,
                               levels = c(0, 1),
                               labels = c("Non-military", "Military")))
```

Counterfactual Prediction (5/6)

```r
library(ggplot2)

# Apply ggplot()
ggplot(p, aes(x = gwf_duration, y = estimate,
              color = as.factor(gwf_military),
              fill  = as.factor(gwf_military))) +
  geom_line(linewidth = 1) +  # Step 1: Draw survival lines
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              alpha = 0.2, color = NA) +  # Step 2: Add confidence bands
  labs(
    x = "Duration (Years)",
    y = "Survival Probability",
    color = "",
    fill  = ""
  )  # Step 3: Clean axis labels and legend
```

Counterfactual Prediction (6/6)



Differences in Predicted Survival Probability (1/3)

- Difference means that:
  - $\mathbb{E}(Survival \mid Military) - \mathbb{E}(Survival \mid Non\_Military)$

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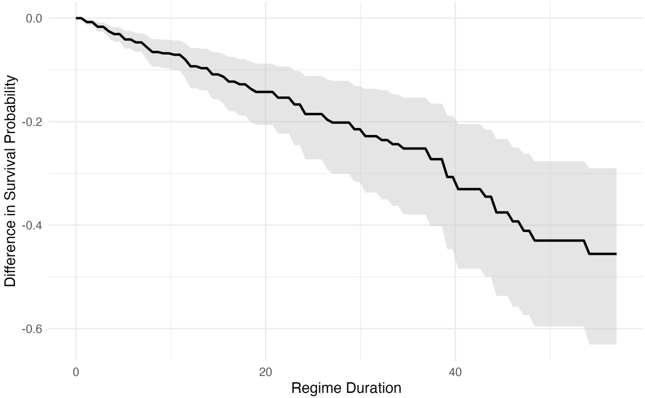
```r
tmax <- quantile(new_df$gwf_duration, 0.95, na.rm = TRUE)        # (1)

# Generate Predictions
m <- comparisons(
  model,
  variables = "gwf_military",                                     # (2)
  newdata = datagrid(gwf_duration = seq(0, tmax, length.out = 100)),  # (3)
  type = "survival",                                              # (4)
  vcov = "rsample"                                                # (5)
)
```

1. Use the 95th percentile of regime duration to avoid extreme outliers.
2. Compare military vs. non-military regimes.
3. Create a grid of duration values from 0 to tmax
4. Predict survival probabilities over time.
5. Use bootstrap (resampling) to estimate more robust confidence intervals.

Difference in Predicted Survival Probability (2/3)

```r
ggplot(m, aes(x = gwf_duration, y = estimate)) +
   geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
               fill = "grey80", alpha = 0.5) +
   geom_line(size = 1, color = "black") +
   labs(x = "Regime Duration",
        y = "Difference in Survival Probability") +
   theme_minimal(base_size = 13)
```

Difference in Predicted Survival Probability (3/3)



Example 2: Veterans' Administration Lung Cancer Dataset

Fit Cox Model with `coxph()`

- Fit Cox Model with `Veterans`
  - Time variable: `Time`
  - Event indicator variable: `Status`
  - Treatment: `trt`

Fit Cox Model with `coxph()`

- Fit Cox Model with `Veterans`
    - Time variable: `Time`
    - Event indicator variable: `Status`
    - Treatment: `trt`

```
# Load and prepare the data
veteran <- survival::veteran
```

```
# Fit Cox proportional hazards model
model_c <- coxph(Surv(time, status) ~ trt, data = veteran)
summary(model_c)
```

Counterfactual Predictions with `marginaleffects()` (1/3)

- Use avg_predictions() to estimate average predicted survival at each time point for each treatment group.
- Also use `vcov = "rsample"` to obtain bootstrapped standard errors.

Counterfactual Predictions with `marginaleffects()` (2/3)

```r
library(marginaleffects)

min_time <- min(veteran$time[veteran$trt == 2], na.rm = TRUE)
max_time <- max(veteran$time[veteran$trt == 2], na.rm = TRUE)

nd <- datagrid(
  trt = 1:2,                                                    # (1)
  time = round(seq(min_time, max_time, length.out = 200)),      # (2)
  grid_type = "counterfactual",                                 # (3)
  model = model_c                                               # (4)
)
```

1. Specify groups to compare (chemotherapy vs. standard)
2. 100 equally spaced points in time between the first (`min_time`) and the last (`max_time`) occurring event time
3. Tells the function to hypothesize specific values of predictors (`trt ==` 1 or 2), regardless of the observed values in the original dataset.
4. Specify the model used to generate predictions.

Counterfactual Predictions with `marginaleffects()` (3/3)

```
ap_surv <- avg_predictions(model_c,
                           type = "survival",
                           by = c("time", "trt"),
                           vcov = "rsample",
                           newdata = nd)
```

```
tail(ap_surv)
```

```
    time trt    estimate    conf.low  conf.high
195  979   1 0.023418828 0.004668523 0.04676991
196  979   2 0.021896712 0.001999662 0.05977180
197  989   1 0.023418828 0.004668523 0.04676991
198  989   2 0.021896712 0.001999662 0.05977180
199  999   1 0.005365124 0.001153153 0.01426689
200  999   2 0.004885817 0.001413887 0.00758126
```

Plot Counterfactual Predictions (1/2)

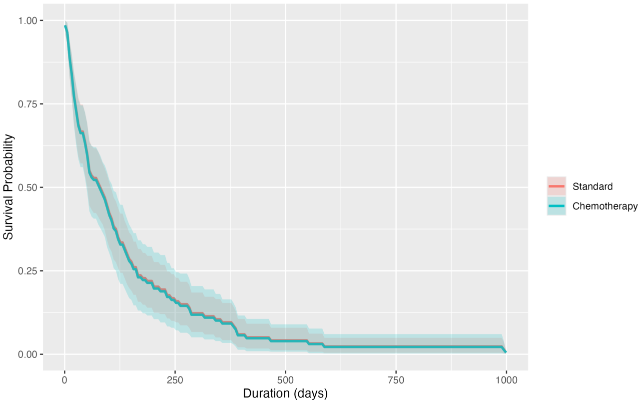
- Similarly, let's factor treatment variable before plotting.

```
# Factoring treatment variable
# library(dplyr)
ap_surv <- ap_surv |>
  mutate(trt = factor(trt,
                      levels = c(1, 2),
                      labels = c("Standard", "Chemotherapy")))
```

Plot Counterfactual Predictions (2/2)

```r
ggplot(ap_surv, aes(x = time, y = estimate,
              color = as.factor(trt),
              fill  = as.factor(trt))) +
  geom_line(linewidth = 1) +  # Step 1: Draw survival lines
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              alpha = 0.2, color = NA) +  # Step 2: Add confidence bands
  labs(
    x = "Duration (days)",
    y = "Survival Probability",
    color = "",
    fill  = ""
  )
```

Plot Counterfactual Predictions



Counterfactual Comparisons with `marginaleffects()`

- We can also compare the average survival probability at different and specific time points. We use `comparison()` here and specify the time points we want in the `datagrid()` argument.
  - We generate a new dataset called newdata using `datagrid(time = c(100, 200, 300, 400, 500))`.
  - Also Specify the `type = "survival"`
    - What does this mean?

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  - Also Specify the `type = "survival"`
    - What does this mean?

```r
# Compare at specific time points
library(marginaleffects)
time_comp <- comparisons(
  model_c,
  variables = "trt",
  newdata = datagrid(time = c(100, 200, 300, 400, 500)),
  type = "survival"
)
```

Counterfactual Comparisons with `marginaleffects()`

```r
time_comp
```

```
 time Estimate Std. Error       z Pr(>|z|)   S   2.5 % 97.5 %
  100 -0.00646     0.0658 -0.0982    0.922 0.1 -0.1355 0.1225
  200 -0.00580     0.0590 -0.0982    0.922 0.1 -0.1215 0.1099
  300 -0.00453     0.0461 -0.0982    0.922 0.1 -0.0949 0.0858
  400 -0.00294     0.0299 -0.0982    0.922 0.1 -0.0616 0.0557
  500 -0.00230     0.0234 -0.0982    0.922 0.1 -0.0482 0.0436

Term: trt
Type: survival
Comparison: +1
```